Entomological Characteristics of Malaria Transmission in Manhiça, a Rural Area in Southern Mozambique

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ABSTRACT From October 1997 to September 1998, an entomological survey was carried out in Manhiça, Mozambique, to describe the anopheline population and intensity of malaria transmission. Ten different huts were randomly selected for entomological surveillance throughout the year. CDC light trap collections were conducted during three nights each month. Additional knockdown spraying catches were carried out in the morning, after the last catch. A total of 17,245 Culicinae and 1,251 Anophelinae were collected during the study. There was substantial house to house variation and seasonality in the distribution of Anophelinae population, with a peak in April towards the end of the warm and rainy season. Four species of genus *Anopheles* (Diptera: Culicidae) were described: *Anopheles funestus* Giles, *Anopheles tenebrosus* Dönitz, *Anopheles arabiensis* Patton, and *Anopheles merus* Dönitz. *An. funestus* constitutes 72.3% of the anopheline population. The estimated sporozoite rate was 1.2% and the average entomological inoculation rate for the area was 15 infective bites per person per year.

KEY WORDS *Anopheles*, EIR, CDC light traps, malaria, Mozambique

TRANSMISSION OF MALARIA IS intense in most parts of Africa south of the Sahara where the highest burden of this disease occurs. *Plasmodium falciparum* malaria is thought to be responsible for 1.5–2.7 million deaths each year, mostly in children under 5 yr of age (WHO 1998). The description of the epidemiology of malaria is often focused on clinical parameters such as prevalence of parasitemia. However, entomological information such as vector species and density, proportion of infected mosquitoes, and the entomological inoculation rate (EIR), are key transmission parameters essential to the understanding of the epidemiology of malaria in an area and the planning of control measures.

The use of light traps in combination with a human being sleeping under a untreated mosquito net has been widely discussed in the last decades as an alternative to the human-biting catch (HBC), especially for endophagic anophelines (Zaim et al. 1986, Service 1993, Smith 1995). Although HBC reflects the fraction of mosquitoes biting humans, it depends on subjective aspects such as the motivation or skill of the collector. Moreover, exposing workers to infective malaria mosquitoes renders HBC ethically unacceptable. Even though in some cases no proportionality was found between CDC traps and human biting collections (Mbogo et al. 1993), many other works showed the capability of the use of light traps in combination with untreated bed nets (Shiff et al. 1995, Davis et al. 1995, Costantini et al. 1998), and this methodology is routinely used in Africa.

Considered to be one of the most important public health problems in Mozambique, malaria is the leading cause of mortality and morbidity. especially in remote areas with little or no access to public health services (Mabunda 1998).

In the southern part of the country, Manhiça District represents a typical rural community where malaria is transmitted all year. More than 90% of these cases are caused by *P. falciparum* (Saute 2004). Several studies reporting malaria and its vectors in the south of Mozambique have been published (Petrarca et al. 1984, Charlwood et al. 1998, Mendis et al. 2000), but mostly from urban or peripheral areas.

As a part of a comprehensive description of the epidemiology in Manhiça district, an entomological survey was designed to obtain basic information for the estimation of the vectorial transmission in Manhiça caused by *P. falciparum*.

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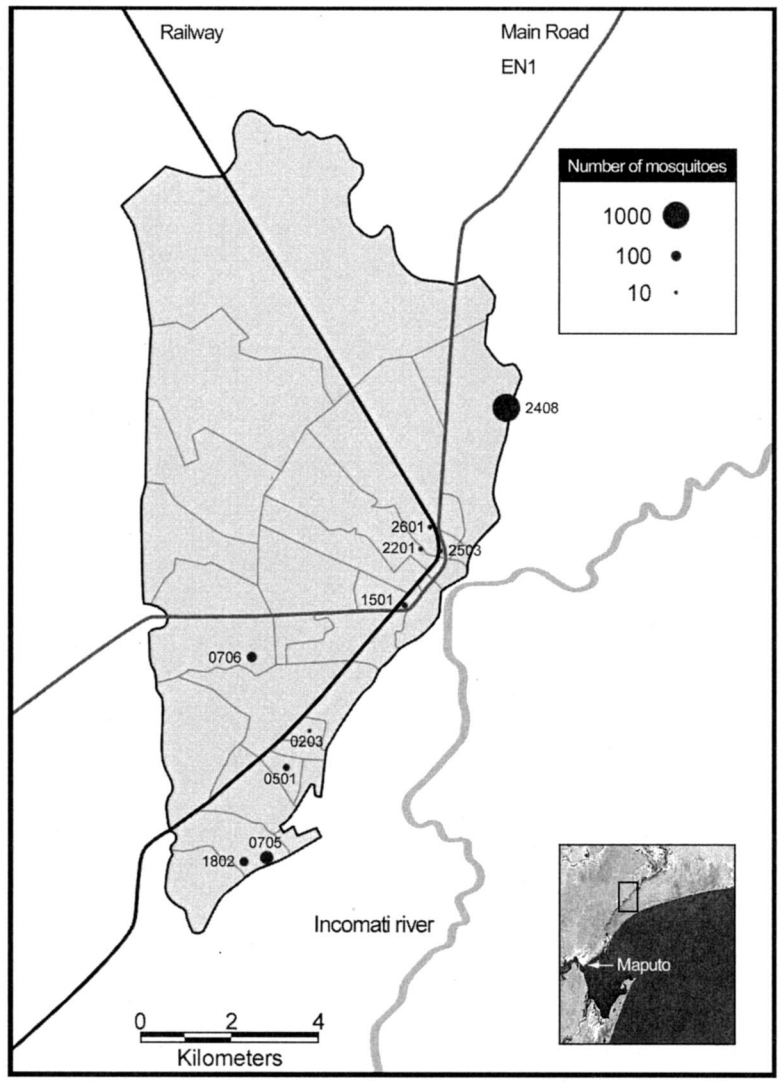


Fig. 1. Map of the Manhiça study area and mosquito collection sites. Dots and corresponding numbers represent the selected houses. The size of the dot is logarithmically proportional to the number of mosquitoes caught.

Materials and Methods

Study Site. The village of Manhiça, 25° 24′ S and 32° 48′ E, is situated in an area of flat bush savanna along the coast of Mozambique, 80 km north of the capital Maputo. The village has a population of 35,000 inhabitants and an approximate extension of 120 km^2. It is located in a well-drained sandy elevated plane, surrounded by the Incomati River meanders (Fig. 1). The lower river floodplain supports anopheline larval breeding all year. Villagers are principally farmers who grow sugarcane, bananas, and rice, although inhabitants also include workers from two large sugar-processing factories. Houses are typically constructed with reeds and have thatched or corrugated roofs. Most of the population inhabits the elevated area, whereas the lower plane, where anopheline breeding sites are located, is used for crops.

The climate is subtropical with two principal seasons, one warm and rainy lasting from October to April–May, and one cold and dry over the remaining months. The annual rainfall was 1,471 mm during the study period. The average temperature was 23.5°C, whereas the mean for July was 19.9°C and for 26 January 26.3°C (Fig. 2).

Mosquito Collections.

Adult mosquitoes were collected monthly between October 1997 and September 1998, by using CDC light traps in combination with human bait sleeping under bed nets, and knockdown collections as described below. To obtain a representation of all area, 10 houses from Manhiça village were randomly selected from the housing database established through the Demo-

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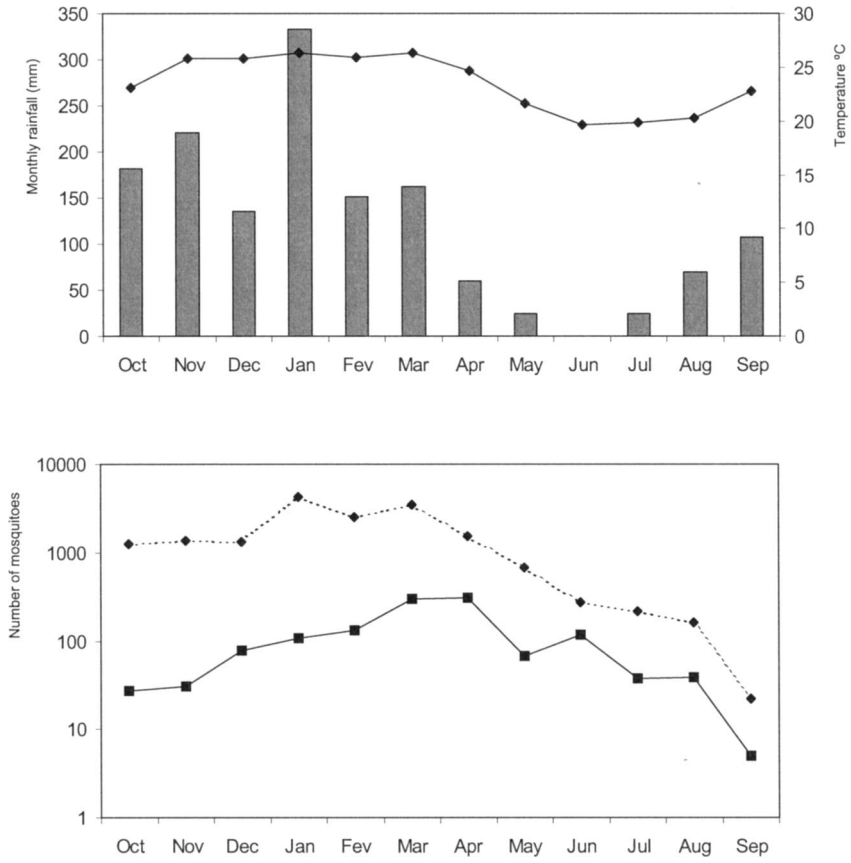


Fig. 2. Meteorological data and captures. Top, rainfall (bars) and monthly mean temperatures (line). Bottom, total number of mosquitoes collected by all methods in a logarithmic scale. The dotted line represents Culicinae and the solid line Anophelinae.

graphic Surveillance System (DSS) (Alonso et al. 2002) in which each house of the village has been georeferenced.

Housing inclusion criteria were only one adult slept in the hut, the house had to be accessible all year around by vehicle, and no insecticide had been sprayed in the months before or during the study. Each month, vectors were sampled as follows: on two consecutive nights during the first week, mosquitoes were collected with the aid of a CDC light trap in each house. In the third week, a CDC light trap was placed in each house for one night, followed by a spray knockdown collection the following morning in the same houses. All selected huts were made with wood and reeds and thatched or metallic roofs.

CDC light traps (model 512, John Hock, Gainesville, FL) were installed before sunset and removed after sunrise. Traps were placed in rooms were a single adult was sleeping under a bed net. The owner was responsible for the proper use of the bed net during the study period. All traps were positioned in the same manner close to the bed net and near the top, to standardize mosquito sample collection (Mboera et al. 1998).

For knockdown collections, mosquitoes were collected using pyrethrin nonresidual spray cans (WHO 1975). In brief, the floor of the room and furniture (when existing) were covered with white sheets, the surfaces of the room were sprayed, and any mosquito falling 10 min after spraying was collected into a plastic container.

Mosquitoes collected in any of the two different techniques (light trap, knockdown) were transported in plastic bags containing moisture to the laboratory for taxonomic identification and storage in silica gel.

Identification of Samples and Laboratory Processing. All mosquito samples were frozen, sorted into Culicinae and Anophelinae, and counted. Culicinae were discarded and Anophelinae were stored in Eppendorff vials with silica gel for further analysis.

Dried specimens were morphologically identified to species with the keys of Gillies and Coetzee (1987), Jupp (1996), and Hervy et al. (1998) and submitted to different procedures. Special care was taken to avoid humidification of samples during handling.

A part of the body, usually a leg and a wing or the head, was used to identify different species of the *Anopheles gambiae* complex by polymerase chain reaction (PCR) carried out after DNA extraction from individual mosquitoes (Collins et al. 1988). Amplification of species-specific restriction fragment length

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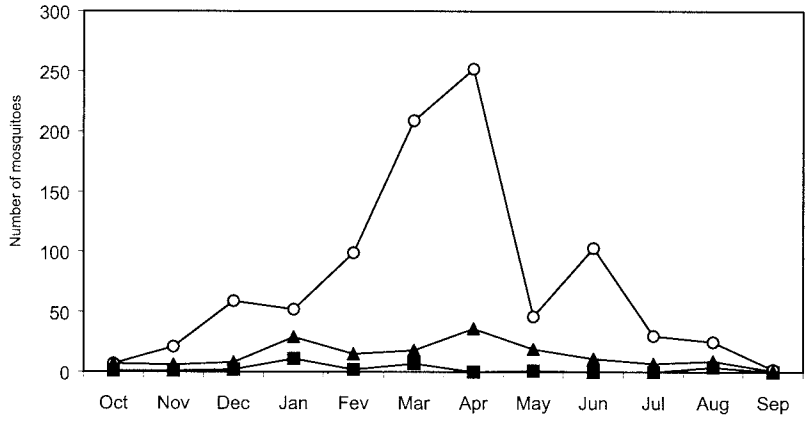


Fig. 3. Monthly densities of different *Anopheles* species. Total numbers of mosquitoes collected with the different methods are represented for different species: *An. funestus* (open circles), *An. tenebrosus* (triangles), and *An. arabiensis* plus *An. merus* (squares).

polymorphisms of the intergenic spacer regions of the ribosomal DNA was performed using universal primers GA (*An. gambiae s.s.*-specific), AR (*An. arabiensis*-specific), and ME (*An. melas*/*An. merus*-specific), according to Scott et al. (1993). Three positive controls (DNA from the different species), two blanks (negative controls from DNA extraction), and one negative control (no template) were used in each PCR reaction. Amplified products were electrophoresed on an ethidium bromide-stained 1% agarose gel and visualized under UV light.

Mosquito blood meals were identified using a two-site enzyme-linked immunosorbent assay (ELISA) (Simão et al. 1995). Antibodies against human, dog, pig, chicken, goat/sheep, cow and rat/mouse G immunoglobulins were used to detect homologous proteins. Four positive controls (homologous blood) and 12 negative controls (heterologous blood) were applied to every plate. The cut-off value of the assay was calculated as the mean plus three times the standard deviation of the negative controls.

An ELISA, using a monoclonal antibody that recognizes a repetitive epitope on the circumsporozoite protein of *P. falciparum* (CSP-Pf), was used to estimate the infection rates in anophelines (Burkot et al. 1984, Wirtz et al. 1987). The ELISA test was carried out following the protocol by Habluetzel et al. (1989), by using two positive and six negative controls on each test plate; the cut-off value was the mean plus three standard deviations of the negative control values. Each test was repeated three times.

Results

A total of 17,245 culicines and 1,251 anophelines were captured between October 1997 and September 1998. Among anophelines, *Anopheles funestus* Giles represented 72.3% followed by *Anopheles tenebrosus* Dönitz (13.3%) and *An. gambiae* s.l. (2.3%). PCR classification of the *An. gambiae* complex showed that it was composed of *Anopheles arabiensis* Patton (96.6%), with 3.4% *Anopheles merus* Dönitz; 12.1% of mosquitoes were not identifiable because of the lack of legs, wings, or other important body parts.

Subfamily Culicinae were represented by different species, but principally *Mansonia africana* (Theobald), *Culex quinquefasciatus* Say, and *Aedes* (*Stegomyia*) sp.

The relationship between mosquito densities with the rainfall and temperature pattern in the area is shown in Fig. 2. Relative abundance and species distribution patterns over the year of study are shown in Fig. 3. *An. funestus* is the most abundant species during all seasons. Its density increases in February, after the peak January rainfall, and reaches its maximum in April, coinciding with the end of the rainy season, and before temperatures decline. Similar patterns occurred for *An. tenebrosus* but at lower densities. *An. gambiae* s.l. presence is concentrated between January and March in very low numbers.

The yield and efficiency of the two collection methods were different. Table 1 shows the species distribution of knockdown spray collections and CDC light traps on the previous night. Knockdown spray collections obtained 15.5% of females, whereas CDC light traps captured 84.5%. All species were more present (percentage ranging from 80.7 to 100) in trap catches especially *An. tenebrosus* that has been found only in this type of samples during all year. *An. merus* was not collected in the subset of these samples.

Table 1. *Anopheles* species collected by the capture method

Species	Spray	%	Trap	%
An. funestus	61	19.3	230	80.7
An. tenebrosus	0	0	43	100
An. arabiensis	1	5.9	16	94.1
Unidentified	6	10.9	49	89.1
Total	68	15.5	338	84.5

Data represent daytime knockdown collections and CDC light trap catches from the previous night.

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Table 2. Total number of mosquitoes collected per house by using different techniques

Mo	House code									
	0203	0501	0705	0706	1501	1802	2201	2408	2503	2601
Oct.	0	0	11	3	0	0	1	12	0	0
Nov.	0	2	6	2	0	0	0	21	0	0
Dec.	0	0	7	1	0	0	0	67	0	2
Jan.	0	4	21	0	2	3	0	75	2	1
Feb.	0	0	3	3	1	0	1	123	0	1
March	0	0	6	14	4	25	2	242	3	1
April	1	2	21	13	2	0	0	272	0	1
May	0	1	13	6	1	0	0	46	1	0
June	0	0	6	0	0	0	0	111	0	0
July	0	0	5	-	0	0	0	32	1	0
Aug.	1	3	7	-	0	0	0	27	1	0
Sept.	0	1	0	-	0	0	0	4	0	0
Total	2	13	106	42	10	28	4	1032	8	6

Results of catches in different huts showed marked differences between them (Kruskal–Wallis, $P < 0.0001$). Although anophelines were collected in all houses, most were caught in house 2408 (Table 2). This house is close to large sugarcane fields and other mosquito breeding sites of the Incomati River floodplain.

Using the *P. falciparum* CS-antigen ELISA, 15 anophelines were positive with optical density (OD) values between 0.209 and 0.834. The average control wells had OD 0.147, standard deviation of 0.007, and cut-off value of 0.169. *An. funestus* was the main malaria vector in the area, whereas there were a few *An. tenebrosus* positives albeit with low OD values. Only collections from houses 2408 and 705 had infective mosquitoes. The overall *P. falciparum* sporozoite rate for mosquitoes collected in traps was 1.2%. Monthly variation of sporozoite rates for house 2408 ranged from 0 to 5%, with two peaks in November and May (Fig. 4). Thirty-eight mosquitoes collected resting in houses could be processed for blood origin; all were positive for human blood.

The EIR, or the number of infective bites per human during the year of study, was calculated as the number of mosquitoes in traps per person per night per year multiplied by the sporozoite rate. The overall EIR for Manhiça was 15. Seasonal distributions of sporozoite rate and EIR for house 2408 where the majority of mosquitoes were collected (82.5%) are shown in Fig. 4. EIR increased between November and June coinciding with the highest mosquito densities and both peaked in April. EIR decreased precipitously in July. The sporozoite rates were always positive between November and June, the second of two peaks almost coinciding with the highest mosquito densities in April at the end of the rainy season. The first sporozoite rate peak in November results in lower EIR due to very low mosquito densities and can be caused by the low number of mosquitoes collected than probably biased rate values.

Discussion

Anopheline populations in Manhiça are present throughout the year, although they are lowest during the cold and dry season. The number of mosquitoes caught in each house strongly varies, depending on hut location.

An. funestus is the most abundant anopheline and the main malaria vector in the area. With the excep-

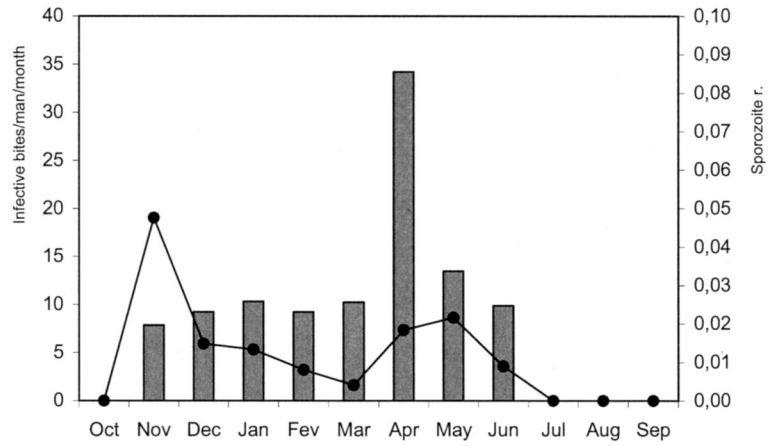


Fig. 4. Sporozoite rate (line) and EIR (bars) per month in house 2408.

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tion of *An. tenebrosus*, species composition of *Anopheles* collected is similar to what was described in the semiurban area of Matola, a neighboring district 80 km away (Thompson et al. 1997, Mendis et al. 2000). The *An. gambiae* complex in the area was composed by *An. arabiensis* and *An. merus*, with no *An. gambiae* s.s., confirming previous finding (Petrarca et al. 1984). *An. funestus* is a mosquito widely distributed in all sub-Saharan Africa (Gillies and De Meillon 1968). In the study area, the number of individuals increased during rainy season but was more abundant at the end of this season, probably due to the stabilization of larval breeding sites consisting in semipermanent clear water collections as grassy swamps, and sugarcane fields.

An. tenebrosus is a large *Anopheles* similar to *An. coustani* living in permanent collections of water with vegetation. Given that it has never been reported as a vector of human malaria (Hervy et al. 1998), and in view of the unusualness of this finding, it would be very interesting to confirm these data on larger samples, possibly using molecular methods. A confirmation of this could not be established neither by PCR or gland dissections due to lack of sample.

Comparing the two sampling methods used, CDC light traps and knockdown spray collections, it was noticed that *An. tenebrosus* has been never found in spray catches probably due to its known exophylic behavior (Gillies and De Meillon 1968). The proportion of *An. funestus* captured by spraying (21%) is higher than with *An. arabiensis* (5.9%) confirming its endophilic tendency (Gillies and Coetzee 1987).

Entomological inoculation rate for the area of Manhiça has been estimated as 15 with sporozoite rate of 1.2%, similar to that estimated for a nearby area (Thompson et al. 1997). Even though anophelines were collected in all area, most were caught in houses near mosquito breeding sites. A recent work reporting human malariometric data from Manhiça (Saute 2004), also suggest that this is an area of perennial transmission with marked seasonality and moderate intensity and reflects the marked heterogeneity in the geographic distribution of the disease being highest in areas close to the flooded plains of the Incomati river where mosquito breeding occurs.

These data contribute to the study of the epidemiology of malaria and provide relevant information for the planning of research studies as well as control programs.

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